TELECOM NETWORK MANAGEMENT SYSTEM (NMS) - MICROSOFT AZURE ARCHITECTURE

Production 3-Tier Enterprise Cloud Architecture | Fully Provisioned via Terraform IaC | Automated via CI/CD Pipelines | Private Connectivity

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MICROSOFT AZURE CLOUD SUBSCRIPTION

1. PRESENTATION & INGRESS TIER

Telecom NOC Operations Users

Network Operations Center Engineers & Staff

- Web Browser Client (HTTPS Port 443)
- Live Telecom Alarms & Metrics Dashboards

HTTPS / TLS 1.3 (TCP 443)

Azure Storage Static Website

Container: Sweb (Public Endpoint)

- Decoupled Single Page Application (SPA)
- HTML5, CSS3, JavaScript UI Bundle
- High durability & serverless auto-scaling
- Low latency delivery for NOC clients

[Verified] Automated artifact sync in CI/CD

IaC & Security Governance

- Provisioned via Terraform azurerm_storage_account
- Static website feature enabled natively in HCL
- Enforced HTTPS-only traffic (TLS 1.2+ minimum)
- 256-bit AES encryption at rest (Microsoft keys)
- CORS policies restricted to authorized endpoints

AZURE VIRTUAL NETWORK (VNet) - 10.0.0.0/16

REST API (HTTPS)

SUBNET: snet-backend-app (10.0.2.0/24) | NSG: nsg-backend

Linux Virtual Machine (Enterprise RHEL / Ubuntu) IP: 10.0.2.10

Nginx Web Server & Reverse Proxy (Ports 80/443)

- SSL/TLS Termination & hardened ciphers
- HTTP rate limiting, CORS headers & Gzip compression
- Access & error log collection managed by systemd
- High-performance proxy pass to FastAPI daemon

Proxy Pass: localhost:8000

FastAPI Telecom Application Server (Uvicorn ASGI)

- Python asynchronous REST API framework
- High-throughput telecom topology ingestion & active alarms
- Managed as enterprise systemd unit (nms-api.service)
- Auto-restart on failure & active health probe (/healthz)
- Async connection pool to Azure PostgreSQL Database

Administered via SSH Keys, RBAC, OS Hardening, Patching & Metrics

Private Link (TCP 5432)

SUBNET: snet-database (10.0.3.0/24) | NSG: nsg-database

Azure Database for PostgreSQL (Flexible Server)

Private Connectivity / VNet Integration (Zero Public IP)

- Dedicated delegated subnet for PostgreSQL Flexible Server
- Private DNS Zone (privatelink.postgres.database.azure.com)
- Stores telecom topology inventory, alarms & system configurations
- Automated backups, point-in-time restore & zone redundancy
- Mandatory TLS/SSL connection encryption enforced
- Credentials secured in Azure Key Vault; monitored in Azure Monitor

Network Security Group (NSG) Strict Enforcement

ALLOW: TCP Port 5432 from 10.0.2.0/24 (snet-backend-app)

DENY: Inbound from Internet / All other subnets

- Zero direct public internet exposure to managed database layer
- Full network isolation satisfying strict telecom enterprise compliance

ENTERPRISE GOVERNANCE, OBSERVABILITY & AUTOMATION SERVICES

Azure Key Vault
Secrets & Credential Mgmt
- DB connection strings & passwords
- Managed Identity RBAC access
- Zero secrets in Git repository

Azure Monitor & Prometheus
Unified Observability
- VM CPU, RAM, Disk & Syslog
- Nginx access & error metrics
- Active NOC telemetry alerting

Terraform IaC Provisioning
Infrastructure as Code
- Declarative Azure provisioning
- Reusable modular architecture
- Checkov & tfsec security scans

Azure DevOps & GitHub Actions
Continuous Delivery
- YAML multi-stage pipelines
- Automated build & unit test
- Multi-env deploy (Dev/Stage/Prod)